

HeatMappers: Fire-Ready Forests Data Challenge

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1 Background

Accurate classification of tree functional types and taxonomic identities is critical for ecological modeling, carbon accounting, and fire behavior simulation. However, Terrestrial Laser Scanning (TLS) datasets, while rich in structural detail, lack species-level labels and functional group identifiers by default. To unlock their full potential, we must infer these biological traits using available features such as tree height, DBH (Diameter at Breast Height), and crown metrics.

1.1 Overall Aim

This task aims to develop machine learning models that enrich TLS-derived tree inventories with ecologically meaningful labels.

1.2 Specific Objectives

Specifically, we pursue the following objectives:

- **Plant Functional Type (PFT) Prediction:** Train a classifier to predict the Plant Functional Type (e.g., evergreen conifer, deciduous broadleaf, shrub) for each tree. Apply this model to the TLS dataset to generate plot-level PFT distributions and compare them to field-observed data.
- **Genus Classification:** Extend the model to predict the genus of each tree (e.g., *Pinus*, *Quercus*). Evaluate whether structural features from TLS can support taxonomic classification at this intermediate level.
- **Species Classification:** Push the model further to predict the full species (e.g., *Pinus ponderosa*). Analyze model performance in light of the challenges posed by limited sample sizes and overlapping morphology among species.

2 Part I: Baseline Model

2.1 Modeling Procedure and Pipeline

Our pipeline began with field-collected tree data from `03.tree.csv`, which included manually recorded attributes such as tree height (HT), diameter at breast height (DBH), and crown base height (CBH), along with scientific species names. We enriched this dataset by joining it with the FIA (Forest Inventory and Analysis) reference species table to map scientific names to species codes (SPCD), genus, and plant functional types (PFT).

Once the biological labels were mapped, we performed data cleaning: standardizing column names, and filtering out incomplete rows to ensure that every training record contained valid HT, DBH, CBH, PFT, genus, and species information.

We trained three types of supervised classifiers on this dataset using `scikit-learn`:

- Random Forest Classifier
- Logistic Regression
- AdaBoost Classifier

Each model used tree height (HT) and DBH as input features and was trained separately for each prediction target:

- PFT (e.g., Evergreen Conifer)
- Genus (e.g., *Quercus*)
- Species Code (SPCD) (e.g., 807 = *Pinus ponderosa*)

The dataset was split using an 80/20 train-test ratio, and model performance was evaluated using accuracy, precision, recall, and F1-score metrics, as well as confusion matrices for interpretability. The Independent variables are Tree height (HT) and Diameter at breast height (DBH).

The figure below visually outlines this end-to-end workflow:

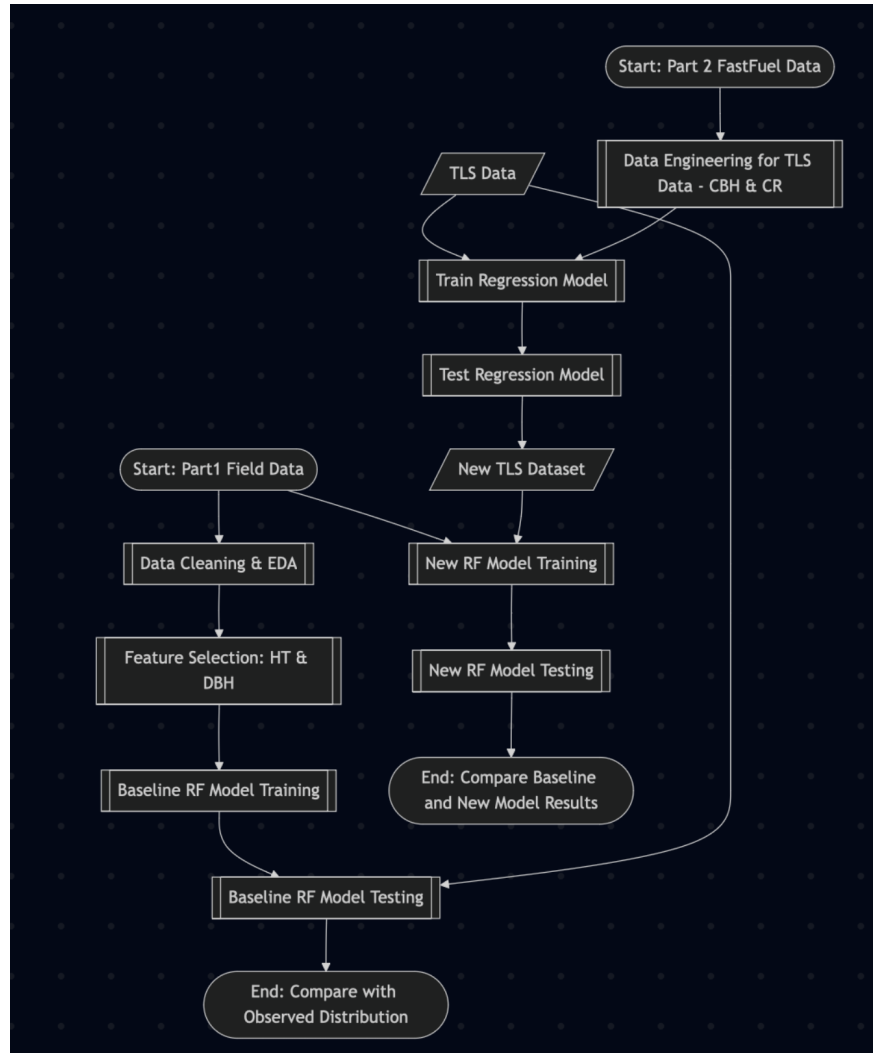


Figure 1: Flowchart comparing the baseline field data approach (Part 1) and the improved FastFuels approach (Part 2).

2.2 Overview of Classification Models We Used

2.2.1 Random Forest Classifier

A robust ensemble model that builds multiple decision trees and averages their predictions. It handles nonlinear relationships and class imbalance well and is resistant to overfitting. We used it since it is a strong baseline for ecological classification tasks due to it being interpretable and having high accuracy. We expected high for dominant classes, slightly weaker for rare categories.

2.2.2 Logistic Regression

A linear model that estimates the probability of each class using a logistic function. Its strengths include simplicity, fast training speed, and ease of interpretation. We used it because it acts as a strong baseline for linearly separable problems and helps benchmark performance against more complex models. Its limitations include potential struggles with complex, nonlinear class boundaries and overlapping features.

2.2.3 AdaBoost Classifier

A boosting model that combines multiple weak learners, often shallow decision trees, into a strong classifier by focusing on difficult-to-classify examples. Its strengths include improving performance on hard-to-predict cases and reducing bias. We used it as it is useful for comparing how boosted ensembles perform against bagging-based models like Random Forest. Its limitations include sensitivity to noisy data and outliers, and potentially less stability with limited training data.

2.3 Data Cleaning Details

To ensure our classification models had access to clean, complete, and labeled data, we performed a structured data cleaning process on the field-collected tree dataset. This involved three main steps: label mapping, column standardization, and filtering incomplete records.

1. Mapping Species Codes and Labels

Each tree in the dataset originally had a scientific species name (e.g., *Quercus kelloggii*), but no numerical species code (SPCD), genus, or plant functional type (PFT). To address this, we mapped scientific names to standardized species codes (SPCD) using the official FIA reference species table. We also joined additional biological labels, including Genus and PFT, based on these species codes. This step allowed us to train models not just to predict numeric identifiers, but to classify trees into ecologically meaningful groups like “evergreen conifer” or “broadleaf deciduous.”

2. Standardizing Variable Names

We then renamed key structural attributes, tree height, diameter at breast height (DBH), and crown base height (CBH), into consistent, easy-to-reference labels (HT, DBH, and CBH). This alignment was important to standardize the field dataset with the naming conventions used in other datasets such as FIA and ALS-derived inventories.

3. Handling Missing Data

Finally, we removed any trees that were missing key features or labels. This included any record that lacked a height, diameter, crown base height, PFT, genus, or species code. Keeping only complete records ensured that our models had all the inputs and targets required for supervised learning, and helped reduce noise in model training and evaluation.

2.4 Initial Approach Summary

Our first modeling strategy focused on building baseline classifiers for Plant Functional Type (PFT), Genus, and Species trained on the high-quality (but short on quantity), field-collected tree data from the `03_tree.csv` dataset. This dataset includes manually measured tree height (HT), diameter at breast height (DBH), and crown base height (CBH), along with taxonomic identifiers such as scientific species name, SPCD, GENUS, and PFT. We then fit the model that we trained using field data onto the TLS dataset, using the same feature space.

2.5 Results

2.5.1 PFT Results

Table 1: **Predicted vs. Observed Plant Functional Type (PFT) Distributions (%)**

PFT	Random Forest (%)	Logistic (%)	AdaBoost (%)	Observed (Field) (%)
Evergreen conifer	85.31	89.58	84.80	71.92
Evergreen broadleaf	12.81	10.42	15.20	20.22
Deciduous broadleaf	1.81	0.00	0.00	6.99
Deciduous	0.07	0.00	0.00	0.87

Interpretation: All models overpredict Evergreen Conifer, with Logistic Regression achieving the highest at 89.58, far above the observed 71.92. The minority classes, particularly Deciduous and Deciduous Broadleaf, are consistently underrepresented, with Logistic and AdaBoost assigning them 0. Random Forest is slightly better than the others at preserving rare classes (we will observe this again later in this report), capturing 1.81 of Deciduous Broadleaf. Overall, Logistic Regression performs best at aligning with the dominant class but fails at minority class representation. Random Forest offers better balance.

2.5.2 Genus Results

Table 2: **Predicted vs. Observed Genus Distributions (%)**

Genus	Random Forest (%)	Logistic (%)	AdaBoost (%)	Observed (Field) (%)
<i>Quercus</i>	16.93	16.35	15.77	24.44
<i>Calocedrus</i>	18.23	19.83	15.85	24.26
<i>Abies</i>	18.45	38.86	2.39	22.07
<i>Pinus</i>	19.83	4.34	61.79	19.06
<i>Pseudotsuga</i>	25.33	20.62	4.20	6.24
<i>Arbutus</i>	0.94	0.00	0.00	2.72
<i>Populus</i>	0.14	0.00	0.00	0.75
<i>Sequoiadendron</i>	0.14	0.00	0.00	0.29
<i>Salix</i>	0.00	0.00	0.00	0.12
<i>Acer</i>	0.00	0.00	0.00	0.06

Interpretation: AdaBoost exhibits significant overfitting to *Pinus* (61.79 predicted vs 19.06 actual) and underpredicts all other categories. Logistic Regression heavily favors *Abies* (38.86 vs 22.07 actual) and *Quercus*, but nearly ignores others like *Arbutus* (0 vs 2.72 observed). Random Forest shows the most balanced genus distribution, but underpredicts dominant genera like *Calocedrus* (18.23 vs 24.26) and *Quercus*. Overall, Random Forest is the most aligned with field observations, although all models struggle with rarer categorizations.

Species Results

Interpretation: Random Forest makes some effort to capture rare species codes (e.g., SPCD 20.0: 2.10 predicted vs 2.60 actual), and maintains broader distribution across classes. Logistic and AdaBoost again show a tendency to cater heavily towards dominant SPCDs: AdaBoost assigns 54.99 to code 15.0, over double the observed value (19.47), with many other classes receiving 0. Logistic Regression assigns 39.51 to SPCD 15.0, ignoring many others entirely. Many true species (e.g., SPCD 312.0, 928.0, 212.0) are completely missed by both Logistic and AdaBoost. Random Forest performs best for species-level prediction due to its ability to capture diversity and rare classes, even if imperfectly. Logistic Regression and AdaBoost show severe bias toward the most common labels, making them poorly suited for fine-grained species classification.

Table 3: **Predicted vs. Observed Species Code (SPCD) Distributions (%)**

	SPCD	Random Forest (%)	Logistic (%)	AdaBoost (%)	Observed (Field) (%)
	81.0	20.41	26.99	18.96	24.26
	15.0	17.66	39.51	54.99	19.47
	801.0	12.59	11.79	7.53	16.35
	108.0	4.56	0.22	1.23	7.51
	116.0	7.09	0.00	8.47	6.82
	202.0	27.21	21.49	8.25	6.24
	818.0	1.88	0.00	0.00	4.16
	122.0	2.82	0.00	0.00	3.12
table-format=2.2	361.0	0.87	0.00	0.00	2.72
	20.0	2.10	0.00	0.00	2.60
	807.0	0.80	0.00	0.00	2.43
	117.0	1.16	0.00	0.58	1.62
	805.0	0.43	0.00	0.00	1.16
	746.0	0.22	0.00	0.00	0.75
	821.0	0.07	0.00	0.00	0.35
	212.0	0.14	0.00	0.00	0.29
	928.0	0.00	0.00	0.00	0.12
	312.0	0.00	0.00	0.00	0.06

2.6 Summary of Baseline Model Results

Across the three classification tasks, Plant Functional Type (PFT), Genus, and Species (SPCD), the results consistently show a decline in prediction accuracy as the classification becomes more specific and granular. This trend is expected: predicting broader categories like PFT (e.g., "Evergreen Conifer") is inherently easier than predicting fine-grained taxonomic labels like genus (e.g., *Pinus*) or species (e.g., *Pinus ponderosa*), especially when using a limited set of input features (e.g., just HT and DBH). PFT classification yielded the highest overall alignment with observed field data, with models achieving high percentages for dominant classes like "Evergreen Conifer." Genus-level classification showed more divergence from field observations, with increasing confusion among similar genera. Species-level classification demonstrated the lowest accuracy, with many rarer species completely missed, especially by Logistic Regression and AdaBoost, which tended to overfit dominant labels. This hierarchy of declining performance reflects both the complexity of the biological taxonomy and the limitations of the feature set used in modeling. Another notable observation across all three classification levels (PFT, Genus, and Species) is that the Random Forest model consistently outperformed Logistic Regression and AdaBoost on low-frequency classes, that is, for tree types with low observed percentages in the field. While Random Forest still predicted small but non-zero proportions for rare entries (e.g., *Arbutus* at 0.94 or SPCD 20.0 at 2.10), both Logistic Regression and AdaBoost often failed to predict these classes entirely, assigning them 0 across the board.

This highlights a key strength of Random Forest: it is better at handling class imbalance and capturing variability, especially when trained on datasets with skewed distributions. In contrast, Logistic Regression and AdaBoost tend to overfit the majority class and neglect minority ones unless explicitly rebalanced.

Another huge limitation with this baseline model is the data we used to train this model. We trained this model on the manually collected field dataset, which merely had 2147 entries. The lack of data makes it difficult to train a model that needs to be able to accurately predict distributions of very specific tree characteristics, such as species and genus.

2.7 Looking Ahead

While the models performed decently, this dataset simply did not contain enough entries for the models to be accurate. Additionally, the lack of data means that the models are highly likely to be overfit meaning that it cannot generalize well to new data. To tackle this issue we decided to repeat the process using the FastFuels dataset and four independent variables instead of two.

3 Part II: Improved Model

3.1 Changing Gears

To improve upon the limitations of the field-trained models, we implemented a second modeling approach using the FastFuels-generated treelist with 545980 entries and 3 additional features: CR (crown radius), CBH (Crown Base Height), and the location site. However, the TLS dataset does not contain CR and CBH, or even location site. We were able to quickly map location site for the TLS dataset using the tree reference dataset, but estimating CR and CBH was a much more difficult task. We decided to take a data engineering approach: implementing a regression model to estimate CR and CBH for the trees in our TLS dataset, which were values we would then use in our feature space for the classification model.

3.2 Overview of Regression Models for Data Engineering

In this improved approach, we needed to impute structural variables not present in the TLS dataset but required for our classification tasks. Specifically, we aimed to estimate CR and CBH using models trained on the field-collected data where these variables were known. Since our FastFuels dataset contains all these values, we trained our models using this data, and later fit them to the TLS data.

3.2.1 Ridge Regression

We selected Ridge Regression for initial imputation attempts. This model is well-suited for high-dimensional or collinear data and provides regularization (L2 penalty) that prevents overfitting. This is particularly important when predicting continuous values like CR or CBH from features like tree height (HT), which might have noise or redundancy, especially when considering different species or site conditions.

3.2.2 Random Forest Regression

In addition to Ridge Regression, we also trained a Random Forest Regressor to predict CR and CBH for the TLS trees. The goal remained the same: generate missing variables using models trained on field data to enable downstream classification. We chose Random Forest Regressor because of it can capture complex non-linear relationships between height and CR / CBH, often better reflecting biological growth patterns. Random Forest is also robust to outliers and captures potential interaction effects between predictors without requiring explicit definition.

3.2.3 Why Both?

By using both Ridge and Random Forest regressors, we could compare a regularized linear approach against a powerful nonlinear ensemble method for variable imputation. This dual-modeling approach helped ensure the imputed features were as accurate as possible, since we would then be using these values in the final classifiers.

3.3 Regression Performance

The regressors achieved the following performance (R-squared) on predicting CBH and CR using the field data as a test set:

- **Ridge Regression:**

- CBH prediction: $R^2 \approx 0.837$
- CR prediction: $R^2 \approx 0.698$

- **Random Forest Regression:**

- CBH prediction: $R^2 \approx 0.993$
- CR prediction: $R^2 \approx 0.995$

Given the significantly higher R^2 values, the Random Forest Regressor models were chosen for the final imputation. We also double-checked this finding by plotting histograms of the actual and predicted distributions of these features, and as we explained earlier, the Random Forest Classifier did a much better job of capturing tails of distributions, whereas ridge regression excelled in areas where the distribution was concentrated. These strong results allowed us to impute CBH and CR using Random Forest Regression with high confidence onto the TLS dataset, and we saved the improved TLS dataset with the imputed values as a .csv file for our last notebook.

3.3.1 Retraining Classifiers with Enriched Data

With the imputed CBH and calculated CR variables added to the TLS treelist, we retrained the same classification models (Random Forest, Logistic Regression, AdaBoost). This time, we used an expanded feature set:

- Height (HT)
- Diameter at Breast Height (DBH)
- Crown Base Height (CBH) (predicted via Random Forest Regressor)
- Crown Ratio (CR) (predicted via Random Forest Regressor)
- Site name (`site_name_cat`) (categorical encoding of plot location)

This richer dataset allowed us to test if including more structural features and site information improved classification performance across the target levels:

- PFT (Plant Functional Type)
- Genus
- Species Code (SPCD)

3.4 Results (FastFuels Approach)

Now that our new dataset, enriched with imputed CBH and CR, was fully prepared, we repeated the classification process used for the initial field-data method. The resulting distributions are shown below compared to the observed field data proportions.

3.4.1 PFT Results

Table 4: Predicted vs. Observed PFT Distributions (%) - FastFuels Approach

PFT	Random Forest (%)	Logistic (%)	AdaBoost (%)	Observed (Field) (%)
Evergreen conifer	80.03	68.31	81.48	71.92
Evergreen broadleaf	12.66	11.65	12.23	20.22
Deciduous broadleaf	6.73	17.29	6.22	6.99
Deciduous	0.58	2.10	0.00	0.87
Evergreen	0.00	0.29	0.07	0.00
Flowering plants	0.00	0.36	0.00	0.00

Interpretation: Random Forest and AdaBoost predict the dominant class Evergreen Conifer quite closely to the observed field proportions (80.03 and 81.48 vs 71.92), though slightly overestimating it. Logistic Regression underperforms on this dominant class with 68.31. Minority classes like Deciduous broadleaf seem somewhat better captured compared to the baseline models; Random Forest assigns 6.73 (vs. 6.99 observed). Extremely rare classes like Evergreen and Flowering plants receive predictions near zero from Random Forest and AdaBoost, which is reasonable given their near-zero presence in the reference field data.

3.4.2 Genus Results

Table 5: Predicted vs. Observed Genus Distributions (%) - FastFuels Approach

Genus	Random Forest (%)	Logistic (%)	AdaBoost (%)	Observed (Field) (%)
<i>Quercus</i>	20.04	27.86	15.12	24.44
<i>Calocedrus</i>	9.84	17.73	8.18	24.26
<i>Abies</i>	13.89	13.39	15.99	22.07
<i>Pinus</i>	34.01	17.37	27.93	19.06
<i>Pseudotsuga</i>	20.26	20.77	32.78	6.24
<i>Arbutus</i>	0.07	0.00	0.00	2.72
<i>Populus</i>	0.00	0.00	0.00	0.75
<i>Sequoiadendron</i>	0.00	0.00	0.00	0.29
<i>Salix</i>	0.00	0.00	0.00	0.12
<i>Acer</i>	1.09	1.01	0.00	0.06
<i>Fraxinus</i>	0.43	0.36	0.00	0.00
<i>Cercocarpus</i>	0.00	0.43	0.00	0.00
<i>Picea</i>	0.07	0.00	0.00	0.00
<i>Lithocarpus</i>	0.29	0.00	0.00	0.00
<i>Prunus</i>	0.00	0.58	0.00	0.00
<i>Umbellularia</i>	0.00	0.51	0.00	0.00

Interpretation: Random Forest distributes predictions more evenly among several major genus, such as *Quercus* (20.04 vs. 24.44 observed) and *Pinus* (34.01 vs. 19.06). AdaBoost seems to favor *Pseudotsuga* (32.78 vs 6.24) and *Pinus*. Logistic Regression inflates *Quercus* to 27.86. Rare classifications (e.g., *Populus*, *Sequoiadendron*, *Salix*) are still significantly inaccurate by all models, but the predictions seem better than the baseline model.

3.4.3 Species Results

Table 6: Predicted vs. Observed Species Code (SPCD) Distributions (%) - FastFuels Approach

SPCD	Random Forest (%)	Logistic (%)	AdaBoost (%)	Observed (Field) (%)	Species Name (from Field Ref.)
81.0	13.46	22.87	21.13	24.26	<i>Calocedrus decurrens</i>
15.0	11.36	13.53	9.77	19.47	<i>Abies concolor</i> (Field SPCD likely)
801.0	7.67	2.68	0.00	16.35	<i>Quercus douglasii</i>
108.0	6.30	0.07	0.07	7.51	<i>Pinus lambertiana</i>
116.0	1.59	0.14	0.00	6.82	<i>Pinus ponderosa</i>
202.0	22.14	19.25	2.46	6.24	<i>Pseudotsuga menziesii</i>
818.0	6.01	22.36	0.00	4.16	<i>Quercus kelloggii</i>
122.0	18.81	14.25	22.79	3.12	<i>Pinus jeffreyi</i>
361.0	0.14	0.00	0.00	2.72	<i>Arbutus menziesii</i>
20.0	0.94	1.01	0.00	2.60	<i>Abies magnifica</i>
807.0	2.03	0.00	0.00	2.43	<i>Quercus garryana</i>
117.0	3.26	0.22	42.84	1.62	<i>Pinus monticola</i>
805.0	1.16	0.00	0.94	1.16	<i>Quercus chrysolepis</i>
746.0	0.00	0.00	0.00	0.75	<i>Populus tremuloides</i>
821.0	0.07	0.00	0.00	0.35	<i>Quercus vaccinifolia</i>
212.0	0.00	0.00	0.00	0.29	<i>Sequoiadendron giganteum</i>
928.0	0.00	0.00	0.00	0.12	<i>Salix</i> sp.
312.0	1.30	1.23	0.00	0.06	<i>Acer macrophyllum</i>
211.0	0.14	0.00	0.00	0.00	(Likely Sequoia sempervirens)
119.0	0.14	0.00	0.00	0.00	(Likely Pinus sabiniana)
127.0	0.65	0.00	0.00	0.00	(Likely Pinus attenuata)
73.0	0.00	0.36	0.00	0.00	(Likely Picea sitchensis)
17.0	0.51	0.00	0.00	0.00	(Likely Abies lasiocarpa)
93.0	0.14	0.00	0.00	0.00	(Likely Libocedrus decurrens - Syn)
109.0	0.00	0.22	0.00	0.00	(Likely Pinus albicaulis)
768.0	0.00	0.43	0.00	0.00	(Likely Quercus wislizeni)
742.0	0.07	0.00	0.00	0.00	(Likely Populus balsamifera)
542.0	0.29	0.36	0.00	0.00	(Likely Lithocarpus densiflorus)
631.0	0.14	0.00	0.00	0.00	(Likely Fraxinus latifolia)
351.0	0.07	0.00	0.00	0.00	(Likely Alnus rubra)
803.0	0.22	0.43	0.00	0.00	(Likely Quercus garryana - Dup?)
811.0	0.07	0.58	0.00	0.00	(Likely Quercus kelloggii - Dup?)
839.0	1.30	0.00	0.00	0.00	(Likely Umbellularia californica)

Interpretation: Random Forest continues to show the best performance by spreading predictions across many SPCD codes, although still struggling with absolute accuracy (e.g., SPCD 81.0: 13.46 predicted vs. 24.26 observed; SPCD 202.0: 22.14 vs. 6.24). AdaBoost exhibits extreme overfitting, assigning 42.84 of predictions to SPCD 117.0 (*Pinus monticola*), far exceeding its observed frequency (1.62). Many rare species codes remain poorly predicted across all models. Logistic Regression again tends to concentrate predictions on a few dominant SPCDs.

3.5 Summary (FastFuels Approach)

Using FastFuels data to train a regression model to engineer a new TLS dataset combined with imputed features (CBH, CR) and site information allowed us to apply the classification framework to a larger, unlabeled dataset, providing scalability. However, this approach also likely introduced other noise or different biases, particularly visible in the Logistic Regression and AdaBoost predictions, potentially due to uncertainties in the imputed features or differences in the underlying tree populations compared to the field plots. Random

Forest consistently maintained reasonable (though imperfect) performance across both the baseline and this improved pipeline, reinforcing its general robustness for such ecological classification tasks.

4 Comparison of Methods

4.1 Plant Functional Type (PFT)

- Accuracy remained relatively high for dominant classes in both approaches.
- Field-trained models (Baseline) might have slightly better captured the exact proportions of dominant vs. rare types seen in the field plots.
- FastFuels models showed potential for better coverage of minority types (e.g., Deciduous Broadleaf), possibly due to the larger dataset.

4.2 Genus Classification

- Both pipelines struggled significantly with rare genus types.
- Random Forest consistently provided the broadest distribution of predictions in both cases.
- Logistic and AdaBoost models showed limited improvement with additional features, continuing to over-predict dominant genus types.

4.3 Species-Level Prediction

- This remained the most challenging task for both approaches, because of how specific of a classification this is.
- Random Forest provided the best spread of predictions across SPCDs in both methods, though still far from accurate.
- AdaBoost and Logistic Regression showed severe overfitting to dominant species in both pipelines, with AdaBoost performing particularly poorly in the FastFuels approach.
- Adding more features (imputed CBH/CR, site) in the FastFuels pipeline showed some significant improvements in PFT and genus classification accuracy - specifically for less common categories - but still struggled for species-level accuracy, suggesting limitations in the predictive power of these features alone or issues with imputed feature quality.

5 Conclusion

Both the field-based baseline and the FastFuels-based improved approach proved viable but serve different purposes:

- **Field-based models:** Offer higher reliability and clarity when high-quality, (and ideally high-quantity) labeled field data is available for training and validation within a specific study area. They are best for understanding model behavior with trusted inputs, because of the accuracy of field-based data collection methods. As mentioned earlier however, there is not enough of this data which leads to insufficient categorizations.
- **FastFuels-based models (with imputation):** Allow the classification process to be extended to larger, unlabeled regions inventoried via remote sensing, making them more practical for landscape-scale ecological modeling, carbon accounting, or fire simulation where field data is sparse.

While our results weren't perfect, we truly believe that our models are still applicable to the overall goal of this project: understanding forest structure to predict fire risk. Even though our proportions weren't exactly perfectly matching up with the field data distributions, an important thing to note is that the field data itself is only a collection of samples from the various sites in the dataset, and it is highly possible that the baseline values (from the field data) that we were comparing our results to are highly inaccurate or unrepresentative of the overall population distribution of trees in the area.

If we were to continue this data-based investigation, we believe that we should integrate different sources of data and primarily focus on developing a better feature space which could be used to predict tree genus and species more accurately. For example, we could integrate ALS characteristics, or other data from the Intellimon source.

6 Contributions & Citations

1. **Ansh Bhatnagar:** Project Manager & Machine Learning Engineer
2. **Jason Qin:** Machine Learning Engineer
3. **Jeronimo Adames-Baena:** Data Scientist
4. **Ishayu Ghosh:** Data Scientist & Report Manager
5. **Special Thanks** to Pedro Antonio Ramonetti Vega and the Data Challenge Team for their guidance and support through this project.
6. **Citation:** We utilized several LLM models, primarily ChatGPT and Claude, to help interpret and ease understanding of topics discussed in Sprints 1-3. We also utilized these agents to proofread our report and debug syntax issues with our code.